

JEE(Main+Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II & III]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,C	A,D	A,B,D	A,B,C	A,B,D	B	D	C	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	4.00	9.00	1.05	5.60	8.36	1.50	1.49		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	B,C,D	B,D	A,B,D	A,B,C	A,C,D	A	C	B	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	4.85	3.00	3.50	6.50	6.00	5.00	4.60	8.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	B,C,D	B,D	B,C	C,D	C,D	C	D	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	7.00	0.33	9.00	19.80	0.50	0.75	2.50		

JEE(Main+Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II & III]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	C	C	A	D	B	B	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	D	C	D	A	B	C	C	C
SECTION-II	Q.	1	2	3	4	5					
	A.	21.00	10.00	6.00	5.00	2.00					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	D	D	C	C	B	D	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	D	D	A	C	D	C	C	B
SECTION-II	Q.	1	2	3	4	5					
	A.	8.00	0.00	2.82 OR 2.83	1.50	162.50					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	B	D	C	B	C	D	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	D	A	D	B	B	A	C	C
SECTION-II	Q.	1	2	3	4	5					
	A.	3.50	4.00	0.20	2.00	1.25					

JEE(Main+Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II & III]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (B,C,D)****Sol.** Plane angle is dimensionless

$$[\omega A] = [v]$$

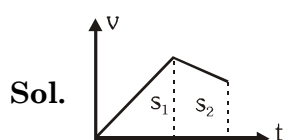
$$[\omega]L = LT^{-1}$$

$$[\omega] = T^{-1}$$

$$\left[\sqrt{\frac{k}{m}} t \right] = 1$$

$$k = \frac{m}{t^2} = MT^{-2}$$

$$\left(\sqrt{\frac{k}{m}} \right) \{t\} = 1 \Rightarrow \sqrt{\frac{k}{m}} = T^{-1}$$

2. **Ans. (A,C)**

For first half a is positive and in direction of velocity

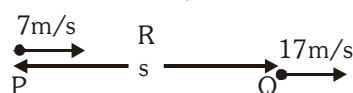
For second half a is negative and in opposite to velocity. From $v-t$ graph $s_1 < s_2$

3. **Ans. (A,D)****Sol.** Distance $\geq$ |Displacement|4. **Ans. (A, B,D)****Sol.** By applying 3rd equation of motion

$$v^2 - u^2 = 2as, 289 - 49 = 2as$$

$$\frac{120}{5} = a, v^2 - 49 = 2 \times \frac{120}{5} \times \frac{5}{2}; v^2 = 169$$

$$\Rightarrow v = 13 \text{ m/s,}$$

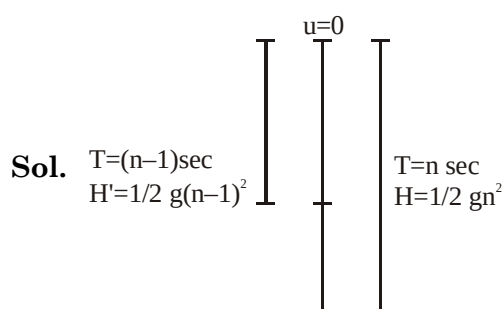


$$\text{Time for P to R} \Rightarrow \frac{13-7}{a} \Rightarrow \frac{6}{a}$$

$$\text{Time for Q to R} \Rightarrow \frac{17-13}{a} \Rightarrow \frac{4}{a}, \frac{t_1}{t_2} = \frac{3}{2}$$

Average velocity

$$\frac{\text{total distance}}{\text{total time}} \Rightarrow \frac{5}{10/a} \Rightarrow \frac{as}{v-u} \Rightarrow 12 \text{ m/s}$$

5. **Ans. (A, B, C)**

Given

$$H - H' = 35$$

$$\frac{1}{2} gn^2 - \frac{1}{2} g(n-1)^2 = 35$$

$$n = 4$$

$$H = \frac{1}{2} g \times 4^2$$

$$= 5 \times 16 = 80 \text{ m}$$

$$v = u + gt$$

$$= 0 + 10 \times 4 = 40 \text{ m/s}$$

6. **Ans. (A,B,D)**7. **Ans. (B)****Sol.** At position P_1 , x increasing $\Rightarrow$ moving in

$+x$ and $\frac{dx}{dt}$ decreasing $\Rightarrow$ speed decreases

8. Ans. (D)

Sol. (7 & 8)

At position P_2 : $\frac{dx}{dt} = 0 \Rightarrow$ instantaneously

at rest

Just after P_2 : x decreases $\Rightarrow$ moves along -ve x -direction.

9. Ans. (C)

10. Ans. (D)

Sol. (9 and 10)

$$x = 3t^2 - 2t$$

$$v = 6t - 2 = 2(3t - 1)$$

$$\text{speed } (v_{\min}) = 0$$

SECTION-II

1. Ans. 3.00

Sol. Average velocity = $\frac{\text{Area under } v-t \text{ graph}}{\text{time interval}} =$

$$\frac{2 \times 3 + 6 \times 1}{4} = 3 \text{ m/s}$$

2. Ans. 4.00

Sol. At maximum velocity :

$$a = \frac{dv}{dt} = 3 - 0.75t^2 = 0 \Rightarrow t^2 = 4 \Rightarrow t = 2$$

$$\text{Second derivative } \frac{d^2v}{dt^2} = -1.5t$$

which is negative at $t = 2$.

Velocity

$$= \int a dt = \int_0^t (3 - 0.75t^2) dt = 3t - 0.25t^3$$

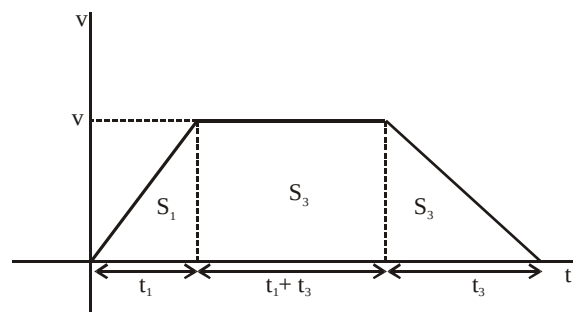
$$\Rightarrow v_{\max} = 3(2) - (0.25)(2)^3 = 6 - 2 = 4$$

3. Ans. 9.00

Sol. If the constant speed attained = v m/s.

$$t_1 = \frac{v}{a} \quad t_3 = \frac{v}{b}$$

$$S_1 = \frac{1}{2}at_1^2 = \frac{v^2}{2a} \quad S_3 = vt_3 - \frac{1}{2}bt_3^2 = \frac{v^2}{2b}$$

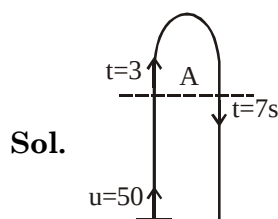


$$S_2 = v(t_1 + t_3) = v\left(\frac{v}{a} + \frac{v}{b}\right)$$

$$\text{avg. speed} = \frac{S_1 + S_2 + S_3}{2t_1 + 2t_3} = \frac{\frac{3}{2}v^2\left(\frac{1}{a} + \frac{1}{b}\right)}{2v\left(\frac{1}{a} + \frac{1}{b}\right)} = \frac{3}{4}v$$

$$v = 12 \text{ m/s, average speed} = 9 \text{ m/s}$$

4. Ans. 1.05



Sol.

Let the height be = h

$$h = u(t) + \frac{1}{2}a(t)^2$$

$$= 50(3) + \frac{1}{2}(-g)(3)^2$$

$$= 150 - 45$$

$$= 105 \text{ m}$$

5. Ans. 5.60

$$\text{Sol. } \frac{dv}{dt} = 6t - 12$$

$$v = 3t^2 - 12t + c$$

$$t = 0, v = 9$$

$$\therefore c = 9$$

$$v = 3t^2 - 12t + 9$$

$$s = \int_0^5 |v| dt = 28$$

6. Ans. 8.36

Sol. 1 calorie = 4.183 = 4.18 $\left[\frac{\text{kg} \cdot \text{m}^2}{\text{sec}^2} \right]$

new system $\Rightarrow 4.18 \left[\frac{1\text{kg}}{2\text{kg}} \cdot \frac{(1\text{m})^2}{(5\text{m})^2} \cdot \frac{(10\text{sec})^2}{(1\text{sec})^2} \right]$

= $4.18 \times \frac{1}{2} \times \frac{1}{25} \times 100 = 8.36 = \text{value of 1 calorie in new system of units.}$

7. Ans. 1.50

Sol. $[Q] = [k][V]^x[F]^y[T]^z$

$ML^2T^{-2} = 1 \times L^x T^{-x} M^y L^y T^{-2y} T^z$

$M^1 L^2 T^{-2} = M^y L^{(x+y)} T^{(-x-2y+z)}$

$y = 1, x + y = 2, -x - 2y + z = -2$

$y = 1, x = 1 \text{ \& } z = 1$

$\therefore (x + y + z) = 3$

8. Ans. 1.49

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A,B,C)
2. Ans. (B,C,D)
3. Ans. (B,D)
4. Ans. (A,B,D)
5. Ans. (A,B,C)
6. Ans. (A,C,D)
7. Ans. (A)
8. Ans. (C)
9. Ans. (B)
10. Ans. (A)

SECTION-II

1. Ans. (4.85)
2. Ans. (3.00)
3. Ans. (3.50)
4. Ans. (6.50)
5. Ans. (6.00)
6. Ans. (5.00)
7. Ans. (4.60)
8. Ans. (8.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B,D)

$y = \frac{x^2 + 2x - 11}{x - 3}$

$\Rightarrow x^2 + x(2 - y) + 3y - 11 = 0$

for real x

$\Rightarrow D \equiv 4 + y^2 - 4y - 12y + 44 \geq 0$

$\Rightarrow y^2 - 16y + 48 \geq 0$

$\Rightarrow (y - 12)(y - 4) \geq 0$

$\Rightarrow y \in (-\infty, 4] \cup [12, \infty)$

$\Rightarrow a = 4 ; b = 12$

2. Ans. (B,C,D)

$(a - 2)x^2 + 2\sqrt{3}x + a > 0$

$a - 2 > 0 \Rightarrow a > 2$

If $D < 0$

$4 \times 3 - 4(a - 2)a < 0$

$\Rightarrow 3 - (a^2 - 2a) - 0$

$\Rightarrow a^2 - 2a - 3 > 0$

$(a - 3)(a + 1) > 0$

$\Rightarrow a \in (-\infty, -1) \cup (3, \infty)$

So common value

$a \in (3, \infty)$

3. Ans. (B, D)

Sol. Let $\log_{10} x = t$

$\therefore t^2 + 2t + 1 = \log_{10}^2 2 \Rightarrow t + 1 = \pm \log_{10} 2$

Hence $x = \frac{1}{20}$ or $\frac{1}{5}$

4. Ans. (B,C)

Sol. $|x - 9| = |9 - x| = 5$

$\Rightarrow x - 9 = \pm 5$

$\Rightarrow x = 14 \text{ or } 4$

sum of roots is 18

5. Ans. (C,D)

Sol. $3 = xy - y - 2x \Rightarrow y(x-1) - 2(x-1) = 5$
 $\Rightarrow (x-1)(y-2) = 5$

$x-1$	$y-2$	(x,y)
5	1	(6,3)
1	5	(2,7)
-5	-1	(-4,1)
-1	-5	(0,-3)

Maximum value of $x+y$ is 9

Minimum value of $x+y$ is -3

6. Ans. (C,D)

Sol. $f(0) = 0, f(1) = 1, f(2) = 2$
 $\Rightarrow f(x) = x$ has roots $x = 0, 1, 2$
 $\Rightarrow f(x) - x = 0$ has roots $x = 0, 1, 2$
 $f(x) - x = (x-0)(x-1)(x-2)$, (Factor)
 Theorem & $f(x)$ is monic polynomial
 $\Rightarrow f(x) = x(x-1)(x-2) + x$
 $f(4) = 28, f(-1) = -7, f(-2) = -26$

Paragraph for Question 7 to 8

7. Ans. (C)

Sol. $\frac{a}{2} < 0 \Rightarrow \boxed{a < 0}$

Min. value = $f(0) = 10$

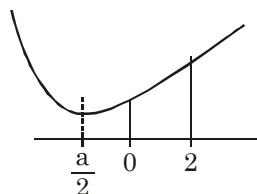
$\Rightarrow a^2 - 2a + 2 = 10$

$a^2 - 2a - 8 = 0$

$\Rightarrow (a-4)(a+2) = 0$

$a = -2, 4$ (Rejected)

$\boxed{a = -2}$



8. Ans. (D)

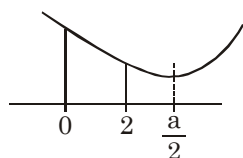
Sol. $\frac{a}{2} > 2 \Rightarrow \boxed{a > 4}$

Min. value = $f(2) = (-7)$

$\Rightarrow 16 - 8a + a^2 - 2a + 2 = -7$

$\Rightarrow a^2 - 10a + 25 = 0$

$\Rightarrow (a-5)^2 = 0 \Rightarrow \boxed{a = 5}$

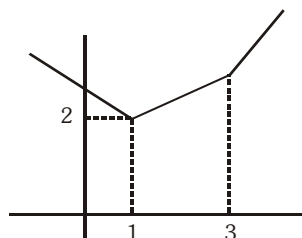


Paragraph for Question 9 to 10

9. Ans. (C)

$f(x) = |x-3| + 2|x-1|$

$$= \begin{cases} -3x+5 & x < 1 \\ x+1 & 1 \leq x \leq 3 \\ 3x-5 & x > 3 \end{cases}$$



$f(x)|_{\min} = 2$

10. Ans. (A)

$g(x) = -2x^2 + 3x - 1$

$g(x)|_{\max} = -\frac{D}{4a} = -\frac{9-4(-2)(-1)}{4(-2)} = \frac{1}{8}$

$f(x)$ & $g(x)$ do not intersect at any point
 $\therefore$ Equation $f(x) = g(x)$ does not have any solution.

SECTION-II

1. Ans. 3.00

$x^4 + x^3 + x^2 + kx^3 + kx^2 + kx + x^2 + x + 1 = 0$

$\Rightarrow (x^2 + kx + 1)(x^2 + x + 1) = 0$

$\Rightarrow x^2 + kx + 1 = 0$

$\Rightarrow D < 0$

$k^2 - 4 < 0$

$-2 < k < 2$

Integral values of $k = -1, 0, 1$.

2. Ans. 7.00

Squaring both sides $64(x^2 + k)^2 = 9x^2$

$(8(x^2 + k))^2 - (3x)^2 = 0$

$(8x^2 + 3x + 8k)(8x^2 - 3x + 8k) = 0$

Real roots are possible when

$D \geq 0$

$\Rightarrow 9 - 256k \geq 0$

$k \leq \frac{9}{256}$

$\Rightarrow \lambda_1 = 9 \text{ \& } \lambda_2 = 256$

$\Rightarrow \sqrt{\lambda_2} - \lambda_1 = 7$

3. Ans. 0.33

$$\frac{\frac{p}{2}(2a_1 + (p-1)d)}{\frac{q}{2}(2a_1 + (q-1)d)} = \frac{p^2 + p}{q^2 + q}$$

$$\Rightarrow \frac{a_1 + \left(\frac{p-1}{2}\right)d}{a_1 + \left(\frac{q-1}{2}\right)d} = \frac{p+1}{q+1}$$

$$\text{Now } \frac{p-1}{2} = 4 \Rightarrow p = 9 \text{ \& } \frac{q-1}{2} = 14 \Rightarrow q = 29$$

$$\Rightarrow \frac{a_1 + 4d}{a_1 + 14d} = \frac{10}{30} = \frac{1}{3}$$

4. Ans. 9.00

Sol. $xy + yz = 18$... (1)

$yz + zx = 20$... (2)

$zx + xy = 8$... (3)

By (1) + (2) - (3), we have

$$2yz = 30 \Rightarrow yz = 15 = 3 \times 5$$

from (1), $xy = 3$

from (2), $zx = 5$

so $xy \times yz \times zx = 3^2 \times 5^2$

$$\Rightarrow \frac{xyz}{xy} = \frac{\pm 15}{xy} \Rightarrow z = \pm 5$$

Case-1 : $z = 5, y = 3, x = 1$

Case-2 : $z = -5, y = -3, x = -1$

So $|x| + |y| + |z| = 9$

5. Ans. 19.80

Sol. $x = (\sqrt{2} + 1)^2 = 3 + 2\sqrt{2}$,

$y = (\sqrt{2} - 1)^2 = 3 - 2\sqrt{2}$,

$$\begin{aligned} x^3 + y^3 &= (x + y)(x^2 - xy + y^2) \\ &= 6((x + y)^2 - 3xy) \\ &= 6(6^2 - 3 \times 1) = 6(33) \\ &= 198 \end{aligned}$$

So $\frac{x^3 + y^3}{10} = \frac{198}{10} = 19.8$

6. Ans. 0.50

Sol. $b, c \in \mathbb{Q}$,

If $3 - \sqrt{3}$ is one root then other root is $3 + \sqrt{3}$

So $-b = \text{sum roots} = 6$

$\Rightarrow b = -6$

$c = \text{product of roots} = 6$

$$\left| \frac{b}{2c} \right| = \left| \frac{6}{12} \right| = 0.50$$

7. Ans. 0.75

Sol. $\sqrt{x+3} = -2x$

L.H.S. $\geq 0 \Rightarrow$ R.H.S. ≥ 0

$$\Rightarrow -2x \geq 0 \Rightarrow x \in (-\infty, 0]$$

for $x \in (-\infty, 0]$

$$(\sqrt{x+3})^2 = (-2x)^2$$

$$\Rightarrow x + 3 = 4x^2$$

$$\Rightarrow 4x^2 - x - 3 = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1+48}}{8} = 1, \frac{-3}{4}$$

$$\Rightarrow x \in \left\{ -\frac{3}{4} \right\}$$

$$\Rightarrow |\alpha| = \frac{3}{4} = 0.75$$

8. Ans. 2.50

Sol. $(x^2 - 1)^3 - (2x - 1)^3 = x^3(x - 2)^3$

$$\Rightarrow (x^2 - 1)^3 + (1 - 2x)^3 + (2x - x^2)^3 = 0$$

Let $a = x^2 - 1, b = 1 - 2x, c = 2x - x^2$

$a + b + c = 0$

$$\Rightarrow a^3 + b^3 + c^3 = 3abc$$

$$\Rightarrow (x^2 - 1)^3 + (1 - 2x)^3 + (2x - x^2)^3$$

$$= 3(x^2 - 1)(1 - 2x)(2x - x^2) = 0$$

$$\Rightarrow x^2 - 1 = 0, 1 - 2x = 0, 2x - x^2 = 0$$

$$\Rightarrow x = \pm 1, x = \frac{1}{2}, x = 0, 2$$

$\Rightarrow$ Sum of all roots of the equation is 2.5.

JEE(Main+Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II & III]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (A)**

Sol. Let the ball is released from height h
 time taken for first bounce/collision.

$$t = \sqrt{\frac{2h}{g}}$$

time taken for n bounce is

$$T = t + 2t + \dots + 2(n-1)t$$

$$= t(1 + 2(n-1)) = 8 \quad \dots(i)$$

Distance in first bounce is h

Distance in n bounce is

$$h + 2h + \dots + 2(n-1)h$$

$$\text{Total distance} = h(1 + 2(n-1)) = 20 \quad \dots(ii)$$

Solving (i) & (ii)

$$h = \frac{5}{4}, n = 8.5$$

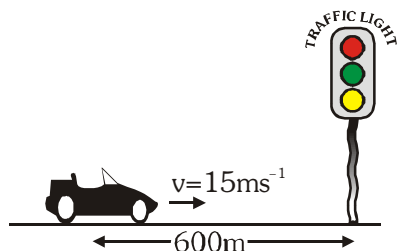
$\Rightarrow 8$ collisions

2. Ans. (C)

Sol. Free falling elevator, there is vacuum in elevator therefore net force on the balloon is $m_{\text{balloon}}g$.

3. Ans. (C)

Sol. Car will come to rest in $t = \frac{v}{a} = \frac{15}{0.3} = 50$ s.



During this time it will cover a distance s

$$= \frac{v^2}{2a} = \frac{15 \times 15}{2 \times 0.3} = 375 \text{ m}$$

Thus required location of car from light
 $= 600 - 375 = 225 \text{ m}$

4. Ans. (C)**5. Ans. (A)**

$$\text{Sol. } \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0 \Rightarrow \vec{F}_3 = -(\vec{F}_1 + \vec{F}_2)$$

$$\Rightarrow \vec{F} = -[(25 \cos 37^\circ \hat{i} + 25 \sin 37^\circ \hat{j}) + (-15 \cos 53^\circ \hat{i} + 15 \sin 53^\circ \hat{j})]$$

$$= -(11\hat{i} + 27\hat{j})$$

6. Ans. (D)

Sol. To have higher average velocity he should move a larger distance on sand.

7. Ans. (B)

Sol. From $v^2 = u^2 + 2as$, $v = \sqrt{2gh}$

$$\Rightarrow v_{B_1} = \sqrt{2g(AB_1)}, v_{B_2} = \sqrt{2g(AB_2)},$$

$$v_{B_3} = \sqrt{2g(AB_3)} \dots$$

If $AB_1, AB_2, AB_3, \dots$ are in G.P. then

$v_{B_1}, v_{B_2}, \dots$ will be in G.P.

8. Ans. (B)

Sol. For uniformly acceleration motion

$$v_{\text{average}} = \frac{v_1 + v_2}{2} = \frac{1 + 3}{2} = 2 \text{ ms}^{-1}$$

9. Ans. (C)

$$\text{Sol. } \beta = [\text{density}]^{1/3} = M^{1/3} L^{-1}, \alpha = [F][M'L^{-3}]$$

$$= M^2 L^{-2} T^{-2}$$

10. Ans. (B)

$$\text{Sol. } 1 = \frac{k\ell}{ma} \Rightarrow \frac{ma}{k} = b = \ell = M^0 L^1 T^0$$

11. Ans. (A)

$$\text{Sol. } 12N = 12(\text{kg m se}^{-2}) = 12 \left(\left(\frac{M}{2} \right) \left(\frac{L}{5} \right) \left(\frac{T}{15} \right)^{-2} \right) \\ = 270 \text{ MLT}^{-2} = 270 \text{ Force}$$

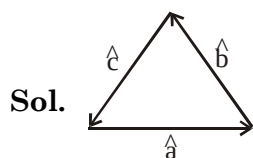
12. Ans. (B)

$$\text{Sol. } v = \frac{dx}{dt} = t^2 + 2t - 3$$

$$\text{when } v = 0 \Rightarrow t = 1 \text{ sec}$$

$$a = \frac{dv}{dt} = 2t + 2 = 2 \times 1 + 2 = 4 \text{ m/s}^2$$

13. Ans. (D)



$$x = \hat{a} \cdot \hat{b} - \hat{b} \cdot \hat{a} - \hat{c} \cdot \hat{a} = -\hat{c} \cdot \hat{a} \\ = -1 \times 1 \times \cos 120^\circ$$

$$\therefore 4x = 4 \times \frac{1}{2} = 2$$

14. Ans. (C)

$$\text{Sol. } <v> = \frac{\text{Displacement}}{\text{Time}} = \frac{\sqrt{2}R}{\frac{\pi R}{2v}} = \frac{2\sqrt{2}v}{\pi}$$

15. Ans. (D)

$$\text{Sol. } \Delta t = \left(\frac{60}{5} - \frac{60}{20} \right) = 9 \text{ sec}$$

16. Ans. (A)

Sol. Velocity is represented by value of v in v - t graph.

17. Ans. (B)

$$\text{Sol. } x = (t^2 - t)(t - 2) \\ = t^3 - 2t^2 - t^2 + 2t \\ x = t^3 - 3t^2 + 2t \\ v = 3t^2 - 6t + 2 = 0$$

18. Ans. (C)

Sol. at point 1

$$v \Rightarrow +v$$

$$a \Rightarrow -ve$$

$\therefore$ slope of v - t graph gives acceleration

at point 2

$$\text{velocity} = 0$$

P Particle change its direction of motion at point 2

$$\vec{a} \cdot \vec{v} > 0 \text{ speeding up}$$

$$\vec{a} \cdot \vec{v} < 0 \text{ speeding down}$$

19. Ans. (C)

$$\text{Sol. use } R = \sqrt{A^2 + B^2 + 2AB \cos \theta} \text{ \&}$$

Displacement = Change in position = final position - initial position

20. Ans. (C)

$$\text{Sol. } x = 36t$$

$$y = 27t - 4.9t^2$$

$$u \cos \theta = 36$$

$$u \sin \theta = 27$$

$$u = 45$$

SECTION-II

1. Ans. 21.00

Sol. The biker wants to meet the cyclist so he will accelerate for some time t & deaccelerate for another time $(t - 1)$ to match his speed 2m/s .

Distance travelled by cyclist

$$S_C = 2(t + t - 1)$$

Distance travelled by biker

$$S_B = \frac{1}{2} 2t^2 \text{ for accelerate.}$$

$$+ 2t(t - 1) + \frac{1}{2} (-2)(t - 1)^2 \text{ for decelerate.}$$

$$S_C + 199 = S_B$$

$$t = 11 \text{ second}$$

$$\text{Total time} = t + t - 1$$

$$= 21 \text{ sec}$$

2. Ans. 10.00

Sol. Area of a-x graph = $\frac{v^2 - u^2}{2}$

so area and velocity will be zero & x = 10

3. Ans. 6.00

Sol. If a ball is thrown up

$$-h = ut_1 + \frac{1}{2}(-g)t_1^2 \quad \dots(i)$$

for down

$$-h = -ut_2 + \frac{1}{2}(-g)t_2^2 \quad \dots(ii)$$

From (i) and (ii)

$$h = \frac{1}{2}gt_1t_2$$

When released

$$h = \frac{1}{2}gt^2$$

$$\text{So } t^2 = t_1t_2 = 9 \times 4$$

$$t = 6 \text{ sec}$$

4. Ans. 5.00

Sol. From the given information,

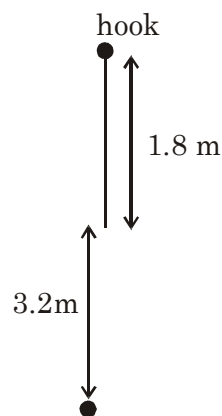
$$n = u + \frac{a}{2}(2n-1)$$

For u to be independent of time,

$$n - an = 0 \Rightarrow a = 1$$

$$\therefore u = \frac{1}{2}$$

5. Ans. 2.00



Sol.

$$3.2 = \frac{1}{2} \times 10 \times t_1^2$$

$$t_1 = \sqrt{0.64} = 0.8 \text{ sec}$$

$$5 = \frac{1}{2} \times 10 t_2^2$$

$$t_2 = 1 \text{ sec}$$

$$\Delta t = 0.2 \text{ sec}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(D)

Sol. IE(M) = E.A(M) and E.A. of Cl is highest

2. Ans.(D)

Sol. $E.N.(x) = \frac{4+13}{2 \times 2.8} \approx 3$; $\overset{3.0}{\text{X}}-\overset{3.5}{\text{O}}-\overset{2.1}{\text{H}} \rightarrow \text{acid}$

$$E.N.(y) = \frac{5.6}{2 \times 2.8} = 1$$
 ; $\overset{1.0}{\text{Y}}-\overset{3.5}{\text{O}}-\overset{2.1}{\text{H}} \rightarrow \text{base}$

3. Ans.(D)

4. Ans.(D)

5. Ans.(C)

Sol. $EN_{\text{Mulliken}} = \frac{IE + E.A}{2}$; $EN(\text{F}) > EN(\text{Cl})$
 $EN(\text{F}) > EN(\text{O})$

6. Ans.(C)

Sol. 89th electron enters '6d' subshell

$$\text{So, } n = 6, \ell = 2 \begin{cases} m = -2 \begin{cases} s = +1/2 \\ s = -1/2 \end{cases} \\ m = -1 \begin{cases} s = +1/2 \\ s = -1/2 \end{cases} \\ m = 0 \begin{cases} s = +1/2 \\ s = -1/2 \end{cases} \\ m = +1 \begin{cases} s = +1/2 \\ s = -1/2 \end{cases} \\ m = +2 \begin{cases} s = +1/2 \\ s = -1/2 \end{cases} \end{cases}$$

7. Ans.(B)

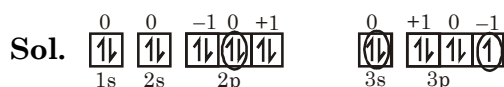
Sol. 8th period will have 50 elements.

Filling of 8s 5g 6f 7d and 8p will occur and last element (inert gas) of this period will be : $118 + 50 = 168$: Z = 168 (Uho)

8. **Ans.(D)**

 9. **Ans.(D)**

 Sol. 2nd electron gain enthalpy is always positive.

 10. **Ans.(C)**

 11. **Ans.(D)**

 Sol. $\text{Li}^{+2} \rightarrow$ single electron : energy : $3s = 3p = 3d$

 12. **Ans.(C)**

Sol. X has 2 valence electrons, hence, belongs to group : 2 (Be, Mg, Ca, Sr, Ba)

 13. **Ans.(D)**

 Sol. Na^+ and Cu^+ are diamagnetic i.e. have no unpaired electrons.

 14. **Ans.(D)**

 Sol. A $\rightarrow$ Pauli, Hund both violated

 B $\rightarrow$ Aufbau violated

 C $\rightarrow$ Pauli, Hund both violated

 D $\rightarrow$ Only Pauli's rule violated

 15. **Ans.(A)**

 16. **Ans.(C)**

 17. **Ans.(D)**

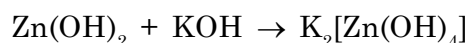
 Sol. I.E. : $\text{Cl}^- < \text{K}^+ < \text{Ca}^{+2}$

 size : $\text{Ca}^{+2} < \text{K}^+ < \text{Cl}^-$

 18. **Ans.(C)**

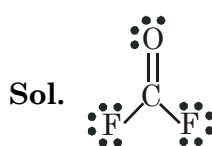
 Sol. $\text{L.E.} \propto \frac{|\text{charge on cation} \times \text{charge on anion}|}{(\text{size of cation} + \text{size of anion})}$

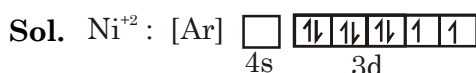
 19. **Ans.(C)**

 Sol. $\text{Zn}(\text{OH})_2$ is amphoteric, hence, can react.

 20. **Ans.(B)**

 Sol. Lattice energy of Al_2O_3 is


SECTION-II

 1. **Ans.(8.00)**

 2. **Ans.(0.00)**

 3. **Ans.(2.82 or 2.83)**


$$\mu = \sqrt{2(2+2)} \text{ BM}$$

 4. **Ans.(1.50)**

 Sol. only acidic oxides : CO_2 , SO_2 , N_2O_5 ($y = 3$)

 Amphoteric oxides : BeO , Al_2O_3 ($x = 2$)

 5. **Ans.(162.50)**

 Sol. $d = r_{\text{F}} + r_{\text{Cl}} - 9(\Delta \text{EN})$
 $= 72 + 99.5 - 9 \times 1 = 162.50$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

 1. **Ans. (D)**

 Sol. $T_{11} + S_{11}$

$$= (a + 10d) + \left[\frac{11}{2}(2a + 10d) \right]$$

$$= 12a + 65d$$

$$= 12 \times 2 + 65 \times 3$$

$$= 219 \text{ Ans.}$$

 2. **Ans. (A)**

Sol. $\log_9 \left[\frac{729}{(3^6)^{1/3}} \right]^{1/4} = \log_9 (3) = \frac{1}{2}$

 3. **Ans. (B)**

 Sol. $\therefore \sqrt{x+2} = x$

$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow x = 2 \text{ or } x = -1 \text{ (reject)}$$

 4. **Ans. (D)**

 Sol. Let the numbers are $a - 3d$, $a - d$, $a + d$, $a + 3d$

 given, $a - 3d + a - d + a + d + a + 3d = 40$

$$\Rightarrow 4a = 40 \Rightarrow a = 10$$

$$\text{and } (a - 3d)^2 + (a - d)^2 + (a + d)^2 + (a + 3d)^2 = 480$$

$$\Rightarrow 4a^2 + 20d^2 = 480$$

$$\Rightarrow 4 \times 10^2 + 20d^2 = 480$$

$$\Rightarrow d^2 = 4 \Rightarrow d = 2, -2 \text{ (reject)}$$

Hence numbers are 4, 8, 12, 16

5. Ans. (C)

Sol. $2 \cdot x^{\log_4 3} + 3^{\log_4 x} = 27$

Put $\{x^{\log_4 3} = 3^{\log_4 x} = t\}$

$\therefore 2t + t = 27 \Rightarrow t = 9$

so, $(3)^{\log_4 x} = 9$

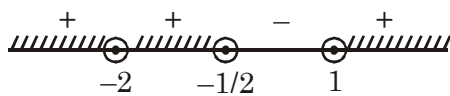
$\Rightarrow \log_4 x = 2$

$\Rightarrow x = 16$

$\Rightarrow \sqrt{x} = 4$

6. Ans. (B)

Sol. $\frac{(x+2)^2}{(2x+1)(x-1)} > 0$



$x \in (-\infty, -2) \cup \left(-2, -\frac{1}{2}\right) \cup (1, \infty)$

7. Ans. (C)

Sol. $\frac{(49)}{(49)^{\log_7 2}} + (5)^{\log_5 \left(\frac{1}{4}\right)}$

$= \frac{49}{(2)^{\log_7 49}} + \left(\frac{1}{4}\right)^{\log_5 5}$

$= \frac{49}{(2)^2} + \left(\frac{1}{4}\right)^1 = \frac{25}{2}$

8. Ans. (D)

Sol. $\therefore \log_2 (9 - 2^x) = x - 3$

$\Rightarrow 9 - 2^x = 2^{x-3}$

$\Rightarrow 9 - 2^x = \frac{2^x}{2^3}$ Put $\{2^x = t\}$

$\Rightarrow 72 - 8t = t$

$\Rightarrow 9t = 72$

$\Rightarrow t = 8$

$\Rightarrow 2^x = 8$

$\Rightarrow x = 3$

9. Ans. (B)

Sol. $x^3 + 0 \cdot x^2 + 2x - 1 = 0$ $\begin{matrix} \alpha \\ \beta \\ \gamma \end{matrix}$

* $\alpha + \beta + \gamma = 0$

* $\alpha\beta\gamma = 1$

$\therefore (\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$

$= (-\gamma)(-\alpha)(-\beta) = -\alpha\beta\gamma = -1$ Ans.

10. Ans. (B)

Sol. $x^2 - 3x + 1 = 0$ $\begin{matrix} \alpha \\ \beta \end{matrix}$

$* \left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + (\alpha^2 + \beta^2)$

$= \left(\frac{\alpha + \beta}{\alpha\beta}\right) + (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(\frac{3}{1}\right) + (3)^2 - 2(1) = 10$ Ans.

11. Ans. (C)

Sol. $\therefore 4x - 1 = \sqrt{5}$

$\Rightarrow 16x^2 - 8x + 1 = 5$

$\Rightarrow 4x^2 - 2x - 1 = 0$

add 2 both side

$4x^2 - 4x + 1 = 2$ Ans.

12. Ans. (C)

Sol. $\therefore x - 2 = 2^{\frac{1}{3}} + 2^{\frac{2}{3}}$ (1)

$\Rightarrow (x - 2)^3 = \left(2^{\frac{1}{3}} + 2^{\frac{2}{3}}\right)^3$

$\Rightarrow x^3 - 2^3 - 3 \cdot x \cdot 2(x - 2) = 2 + 2^2 + 3 \cdot 2^{\frac{1}{3}} \cdot 2^{\frac{2}{3}} \left(2^{\frac{1}{3}} + 2^{\frac{2}{3}}\right)$

$\Rightarrow x^3 - 6x^2 + 12x - 8 = 6 + 6(x - 2)$

$\Rightarrow x^3 - 6x^2 + 6x = 2$

13. Ans. (D)

Sol. $\therefore (a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$

$\Rightarrow (11)^3 = a^3 + b^3 + c^3 + 3 \times 11$

$\Rightarrow a^3 + b^3 + c^3 = 1331 - 33$

$= 1298$ Ans.

14. Ans. (A)

Sol. $\therefore \frac{2x^2 - 4x + 5}{D < 0} = 0$

So, both roots are imaginary & imaginary roots always occur in pair of conjugate.

$\therefore \begin{matrix} x^2 + px + q = 0 \\ \& 2x^2 - 4x + 5 = 0 \end{matrix} \left. \begin{matrix} \text{have both} \\ \text{roots in} \\ \text{common} \end{matrix} \right\}$

$\Rightarrow \frac{1}{2} = \frac{p}{-4} = \frac{q}{5} \Rightarrow p + q = -2 + \frac{5}{2} = 0.5$

3. Ans. 0.20

Sol. $A = \log_3[4(\cos^2(x) + \sin^2(x) - 1)] = \log_3 3 = 1$

$$B = \log_{\sqrt{2}-1} \sqrt{2} + 1 = \log_{\frac{1}{\sqrt{2}+1}} \sqrt{2} + 1 = -1$$

$$C = (2)^{\log_6 6 \times 3} (3)^{\log_6 3} = (2)^{1+\log_6 3} (3)^{\log_6 3} \\ = (2)^1 \cdot (2)^{\log_6 3} \cdot (3)^{\log_6 3} = 2 \cdot (2 \times 3)^{\log_6 3} = 2 \times 3 = 6$$

$$\therefore \frac{A}{B+C} = \frac{1}{-1+6} = 0.20 \text{ Ans.}$$

4. Ans. 2.00

Sol. $\underbrace{\sqrt{(x+7)(x-2)}}_{\text{Non-ve}} + \underbrace{|(x+6)(x-2)|}_{\text{Non-ve}} = 0$

So, $\sqrt{(x+7)(x-2)} = |(x+6)(x-2)| = 0$

$$\therefore x = 2$$

5. Ans. 1.25

Sol. $\Rightarrow (x)^{\log_2^2(x)+4\log_2(x)+3} = (x)^{\log_2(x)+1}$

$$\Rightarrow \boxed{x=1} \text{ or } \log_2^2(x) + 4\log_2(x) + 3 = \log_2(x) + 1$$

Put $\log_2(x) = t$

$$\Rightarrow t^2 + 3t + 2 = 0$$

$$\Rightarrow t = -1 \text{ or } -2$$

$$\Rightarrow \log_2 x = -1 \text{ or } -2 \Rightarrow \boxed{x = \frac{1}{2} \text{ or } \frac{1}{4}}$$

$$\therefore \left. \begin{array}{l} \alpha = 1 \\ \beta = \frac{1}{2} \\ \gamma = \frac{1}{4} \end{array} \right\} \alpha + \beta - \gamma = 1.25$$